

Abrar Ul Riyaz

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SUMMARY

- Backend and AI Engineer specializing in **agentic AI systems, LLM applications, and scalable backend architectures**.
- Doing **ML Research and backend development** at **GAASH Research Lab, NIT Srinagar**, working on GPU-scale model training and vision-language research.
- Experienced in building **production AI systems** using **Python, Django, LangChain, and modern LLM tooling**.

EXPERIENCE

GAASH Research Lab National Institute of Technology (NIT) Srinagar

Apr. 2025 – Present

Research & Backend Development

Srinagar, India (Onsite)

- Conducting applied research in **Machine Learning, LLM reasoning, and Agentic AI systems**, focusing on building autonomous and tool-using AI workflows.
- Training, fine-tuning, and evaluating deep learning models on **NVIDIA H100 GPUs**, optimizing performance and scalability.
- Designing and developing **agentic pipelines (RAG, tool-calling, multi-step reasoning)** using modern LLM frameworks.
- Leading backend development for AI-driven applications, handling **database design, REST APIs, and scalable system architecture**.
- Built and deployed **LLM-powered full-stack systems** using **Django, PostgreSQL, and vector databases**, ensuring reliability and maintainability.

AI4BHARAT Lab Indian Institute of Technology (IIT) Madras

Dec. 2024 – Mar. 2025

Research Intern

Remote

- Conducted **Optical Character Recognition (OCR)** research on 3 Indian Languages: **Urdu, Bengali, and Assamese**.
- Worked with multiple OCR engines, including **AI4Bharats Surya OCR** and **Google Cloud Vision OCR**, for multilingual text extraction.
- Performed a detailed **comparative evaluation** of OCR engines based on accuracy, robustness, and language-specific performance.
- Documented experimental results and contributed to **dataset preparation, benchmarking reports, and performance analysis**.

TECHNICAL SKILLS

Programming Languages : Python, C, C++, Java, JavaScript, TypeScript

Backend & APIs : Django, DRF, FastAPI, RESTful APIs, Pydantic, LangChain, LangGraph, LangSmith, MCP, RAG

LLMs & Model Platforms : OpenAI API, Google Gemini, Anthropic (Claude), Hugging Face (Open-Source LLMs), Ollama

Databases : PostgreSQL, MySQL, MongoDB, ChromaDB, Pinecone

Frontend : HTML, CSS, React, Tailwind CSS, Bootstrap, Streamlit

Machine Learning & AI : Machine Learning, LLMs, Agentic AI, Reasoning Models, NLP, Computer Vision, Generative AI

ML & Data Libraries : NumPy, Pandas, Scikit-learn, Matplotlib, OpenCV

DevOps & Systems : Docker, Kubernetes, Kafka

Core CS : Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

Tools & Platforms : Git, GitHub, Hugging Face, Postman, REST Client, Thunder Client

Deployment : Netlify, Vercel, Render

PUBLICATIONS

Grounded Surgical Benchmark for Language-Conditioned Tool Segmentation [GitHub](#) [arXiv](#)

- **Conference:** MICCAI 2026 (Under Review)
- Introduced a **grounded vision-language benchmark** for surgical scene understanding, enabling models to localize and segment surgical tools from **natural language descriptions**.
- Constructed a **multi-procedure dataset** with **600+ surgical images and 1000+ tool-level annotations**, combining pixel-level segmentation, bounding boxes, and center-point grounding signals.
- Benchmarked multiple **vision-language models** (Qwen-VL, Gemma, InternVL, VisionReasoner) to evaluate **language-conditioned surgical tool localization and segmentation**.

QTrack: Query-Driven Reasoning for Multi-modal Multi-Object Tracking [GitHub](#) [arXiv](#)

- **Conference:** ECCV 2026 (Under Review)
- Proposed a **query-driven tracking paradigm** that formulates multi-object tracking as a **language-conditioned spatiotemporal reasoning problem**.
- Developed **RMOT26 benchmark**, a large-scale dataset with **grounded natural language queries** and temporally consistent annotations for evaluating reasoning-aware tracking.
- Introduced **QTrack**, a reinforcement learning-based vision-language framework with **TAPO optimization** to ensure motion consistency and identity preservation.
- Achieved **state-of-the-art performance** on RMOT26, outperforming strong baselines in both **spatial accuracy and temporal consistency**.

PROJECTS

PromptIQ [GitHub](#)

- **Tech Stack:** Django, REST APIs, React, PostgreSQL, LLM APIs (GPT-4, Gemini)
- Designed and developed a **full-stack AI chatbot** using Django and React, enabling dynamic selection and interaction with multiple **large language models**.
- Implemented **PDF ingestion and document-based question answering**, allowing users to perform natural language queries and generate grounded AI-driven responses.
- Built **context-aware conversational memory** to support multi-turn interactions and maintain coherent, stateful chat history across user sessions.

BIS-Basha [GitHub](#)

- **Tech Stack:** Django, React.js, Sarvam API (Translation), GPT (Summarization), Google Cloud Text-to-Speech
- Developed an **AI-powered multilingual web application** supporting **translation and text-to-speech (TTS)** for **22 Indian languages**, including Hindi, Tamil, and Bengali.
- Implemented **document translation with structure preservation**, retaining original layout elements such as **tables, figures, headings, and formatting** after translation from English to Indian languages.
- Integrated **Sarvam APIs** for high-quality **machine translation** and **Google Cloud Text-to-Speech** for natural-sounding speech synthesis across multiple languages.
- Implemented **AI-driven PDF summarization** using **GPT-based models**, enabling fast and accurate understanding of uploaded documents.
- Designed a **modular and scalable backend architecture** in Django to orchestrate translation, summarization, and TTS processing pipelines.

RagFirst [GitHub](#)

- **Tech Stack:** React (TypeScript), Django, LangChain, Pydantic, OpenAI GPT-4, Vector Embeddings
- Developed a **retrieval-augmented generation (RAG)** application for **document and video understanding**, enabling grounded and context-aware AI responses.
- Implemented **multi-document PDF ingestion** with **semantic search** over vector embeddings, providing accurate AI-generated summaries and information retrieval.

- Built a **context-aware conversational chatbot** supporting **multi-turn question answering** using conversation history and retrieval context.
- Integrated **YouTube video analysis** by generating concise summaries and enabling **natural language Q&A** over video transcripts.
- Designed an **end-to-end retrieval and reasoning pipeline** using LangChain and Pydantic to ensure structured inputs, grounded outputs, and reliable response generation across heterogeneous data sources.

DockVault [GitHub](#)

- **Tech Stack:** Django, REST APIs, PostgreSQL, AI/LLM APIs
- Developed a **secure healthcare data management platform** enabling doctors and healthcare professionals to store and manage **patient records, visits, treatments, and test results**.
- Implemented patient history aggregation to generate **AI-assisted clinical summaries**, highlighting key medical details and potential gaps in prior treatments.
- Designed **AI-driven recommendation support** to suggest possible next steps in patient care based on historical data and established best practices.
- Built modular backend services with **role-based access control** to ensure data integrity, privacy, and secure access to sensitive healthcare information.

EDUCATION

National Institute of Technology (NIT) Srinagar

Nov. 2022 – Jul. 2026

Bachelor of Technology in Information Technology

Srinagar, India

- **CGPA:** 8.27 / 10
- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Machine Learning

Nigeeen Higher Secondary School

Dec. 2019 – Mar. 2021

Higher Secondary (HSC / 10+2)

Srinagar, India

- **Percentage:** 94.6%
- **Subjects:** Mathematics, Physics, Chemistry

LANGUAGES

Languages : English (Professional Proficiency), Urdu (Native), Hindi (Native)